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Order batching in walk-and-pick order picking systems

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This paper investigates the order-batching problem both as a theoretical problem, including its special variants, and as a practical problem, with possible heuristic solution procedures. It shows that certain special cases or variants of the order batching problem are solvable, while other cases of the problem are NP-hard. It also develops a new Mixed-Integer Programming (MIP) model to obtain near-exact solutions to the problem. The MIP provides good lower and upper bounds to the problem, which prove useful in the investigation of the heuristic solution procedures. The paper identifies and modifies slightly the order batching heuristic, which shows the strongest results in the numerical experiments. Although good solutions are obtained with this particular heuristic, the results indicate that it is computationally intensive. Furthermore, solutions obtained through the MIP model suggest there is still room to obtain even better solutions to the problem via heuristic procedures, although the challenge is to do so without further increasing the computational burden.

Keywords: Warehousing; Order picking; Order fulfilment; Order batching

1. Introduction

Unlike commercial distribution centres that perform many unit load picks, retail distribution centres (RDCs) are often faced with a large number of less-than-unit-load picks. Although a few RDCs have been in operation back when customers ordered goods from a catalogue via the telephone, a large number of orders are now processed through many RDCs due to a growing number of customers who shop on the Internet. In fact, it is not uncommon for RDCs to receive a large variety of orders each day with a wide range of types of items ordered. Furthermore, each type of item is often ordered in quantities of only one or two.

Various methods—ranging from manual to automated—are in use today to perform less-than-unit-load picks, which generally fall under ‘order picking’ or ‘order fulfilment’. See Tompkins *et al.* (2003, pp. 268–278, 447–457) for a review of basic order picking methods and equipment. Two fundamental methods that are common are known as ‘operator-to-stock’ and ‘stock-to-operator’ (Tompkins *et al.* 2003, pp. 269, 271). Operator-to-stock methods include walk-and-pick systems using

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picking carts (some of which are computerized ‘smart carts’), order picker trucks, and person-on-board automated storage/retrieval machines. Stock-to-operator methods include carousels, miniload automated storage/retrieval machines, vertical lift modules, and automatic dispensers. The primary goal in the above systems is to pick orders accurately and efficiently before they are sorted/packed and shipped for delivery to the customer, often by a specified due date. To track items or orders, and to improve accuracy as volumes increase, many RDCs deploy technologies such as bar code identification, radio frequency terminals, and pick-to-light systems (Tompkins *et al.* 2003).

The present paper focuses on walk-and-pick systems and studies the problem of batching a set of orders, i.e. a ‘wave of orders’, into order picking trips. The problem was studied previously by Koster *et al.* (1999), Ruben and Jacobs (1999), and Gibson and Sharp (1992). The objective, which we define as the ‘minisum’ objective, is to minimize the total time to pick all the orders in a wave, given finite-capacity pickers. The RDC is assumed to have a simple ‘ladder structure’ in the order picking (or ‘forward’) area, as shown in figure 1. Note that cross-aisles are located only at the front and rear of the forward area. The pick-up/deposit (P/D) point, where each trip starts and terminates, is located at the midpoint of the front cross-aisle, as shown in figure 1. For each trip, the picker is assumed to follow the traversal strategy (Ratliff and Rosenthal 1983, Hall 1993) under which the picker completely traverses, from one end to the other, each aisle that he/she enters. Of course, only those aisles that contain one or more picks are entered. As the picker picks the items, he/she places them in a pick cart (or pick truck). Order integrity can be maintained by placing each item in the appropriate container or compartment in the pick cart/truck.

Hall (1993) showed empirically that the traversal strategy performs well if the number of picks/aisle for a given trip is greater than approximately 3.9 picks/aisle. In practice, the traversal strategy is easy to implement, and in some cases it may be necessary due to physical constraints such as the size and manoeuvrability of the pick cart and/or the width of the aisles.

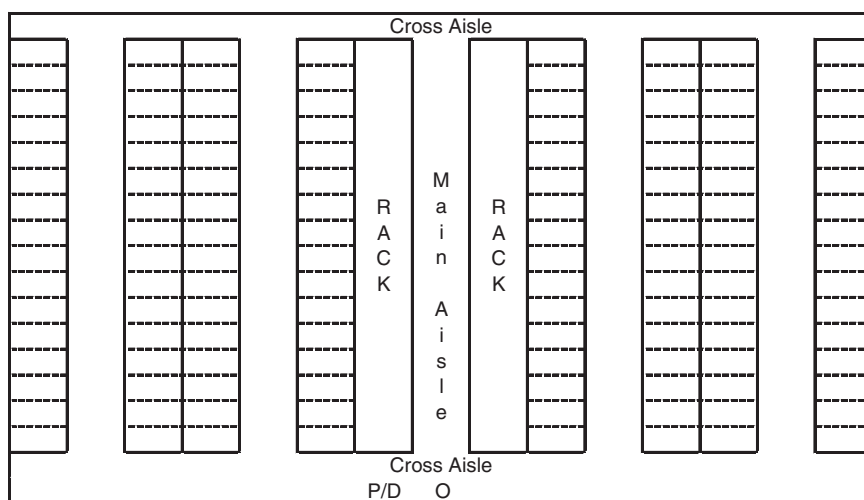


Figure 1. Ladder structure warehouse.

The order batching problem starts with a wave of orders, with each order completely described by the number of line items in the order, and the location of each line item in the forward area. In order picking, a 'line item' corresponds to a Stock Keeping Unit (SKU), which serves essentially as a part number. Henceforth, we refer to a 'line item' as an 'item'. The goal is to separate the orders into batches (or trips), such that the total trip time (which is defined by walking time) to pick all the orders is minimized, subject to two constraints. First, the total size of each batch (or trip) cannot exceed the capacity of the pick cart/truck, which is expressed in items. Second, in order to maintain order integrity, no order can be split over two or more trips. The total time to pick all the orders is simply equal to the sum of the trip times associated with each batch (or trip). With constant picker speed in the forward area, minimizing total distance travelled is equivalent to minimizing total time since time spent actually picking the items from the rack is constant for a given wave of orders. For convenience, in the remainder of the paper, we will use time or distance depending on context.

An alternative objective defined for the order batching problem, say, 'minimax', is based on minimizing the longest trip time among the resulting batches. If the orders are picked *concurrently*, and one is more concerned with passing the *entire wave* of orders to the next process in the shortest possible time, then minimizing the maximum trip time would be desirable. However, doing so also implies that one or more pickers, in general, will have to remain idle while the current wave is being completed. In contrast, if the orders are picked *sequentially*, and each order can be passed immediately to the next process such as packing or shipping without having to wait for the entire wave to be processed, it would be more desirable to use the minimus objective, which amounts to minimizing the *total time* required to process the wave. Depending on the application, we believe either objective can be used. In this paper, as in Koster *et al.* (1999) and Ruben and Jacobs (1999), among others, we focus on the minimus objective. The reader may refer to Gademann *et al.* (2001) for the minimax objective.

2. Literature review

Order batching problems received a fair amount of attention. Koster *et al.* (1999), Rosenwein (1996), Ruben and Jacobs (1999), and Gibson and Sharp (1992) developed heuristics for the minimus problem using the traversal strategy in a forward area with a simple ladder structure. For the same type of forward area, Gademann *et al.* (2001) solved the order-batching problem with the minimax objective.

In their study, Gibson and Sharp (1992) considered Euclidean, rectilinear, Chebyshev, and aisle travel (i.e. traversal) in the forward area. They also considered uniform versus skewed item location, fixed and variable order size, and discrete versus continuous representation of the forward area. Orders are batched to maintain batch sizes as close to 50 items as possible, and the P/D point is located at the left end of the front cross-aisle. The three heuristics the authors study are the four-dimensional space-filling curve, Sequential Minimal Distance, and first-come, first-served as a baseline. For a ladder structure with aisle traversal, they show that the Sequential Minimal Distance heuristic is preferable. With this heuristic, an order

is selected as a seed order and assigned to a batch. Next, the distance from each item in the seed order to the closest item in a candidate order is calculated. Summing up the above distances yields the distance between the two orders. The process is repeated for each unassigned (i.e. candidate) order. The order(s) with the shortest distance(s) to the seed order is/are then added to the current batch until the picker capacity is reached. If any unassigned orders remain, a new seed order is selected to start a new batch.

Rosenwein (1996) also developed heuristics for the minisum problem. Like Gibson and Sharp (1992), the author used a ladder structure with a corner P/D point. The pickers follow the traversal strategy with a capacity given in number of items. Items are located throughout the forward area by order frequency, with those items ordered most often located near the P/D point. The heuristics examined include the Minimum Additional Aisle, Centre of Gravity, and the Gibson and Sharp (1992) Sequential Minimal Distance as a baseline. Rosenwein showed that the Minimum Additional Aisle heuristic outperformed the others in terms of number of tours needed and total distance travelled.

The Minimum Additional Aisle heuristic begins with calculating a score for each pair of orders; it is equal to the minimum of {(additional aisles to pick orders i and j together above order i alone), (additional aisles to pick orders i and j together above order j alone)}. Given the above scores, the pair with the lowest score that does not violate the capacity constraint are removed from the list of orders and combined into a new order, which is added back to the list of orders. In case of a tie, the pair with the largest combined size is selected. Subsequently, scores are recalculated for all pairs of orders on the updated list, and the process is repeated until no set of orders can be further combined due to the capacity constraint. Each set of orders is then picked as a single tour.

As in previous studies, Ruben and Jacobs (1999) used a ladder structure with the traversal strategy, and a picker capacity given in number of items. No P/D point was used; each tour starts at the first item to be picked and ends with the last. For random, turnover-based, and family-based item locations, the authors develop five heuristics: Random Batching, First-Fit-Decreasing, First-Fit-Envelope Based Batching, First-Fit-Class Based Batching, and Sequential Minimal Distance from Gibson and Sharp (1992). Of these, the First-Fit-Envelope Based Batching described in section 3.4 was determined to be the best for the minisum objective.

Koster *et al.* (1999) performed an extensive study of heuristics for the minisum objective. They consider a forward area with pickers of finite capacity, which is expressed in number of items. Pickers follow either the largest gap or the traversal strategy. Items are located uniformly throughout the forward area, and the P/D point is in the corner. The authors separate the batching heuristics into three groups: simple (First-Come-First-Served), seed, and savings. The seed algorithms include Gibson and Sharp's (1992) heuristic as well as Rosenwein's (1996) Minimal Additional Aisle and Centre of Gravity heuristics. The savings algorithms include the Clarke and Wright algorithm (C&W(ii)) based on time savings (see section 3.4). Of these, the C&W(ii) seems to be the best for the minisum objective in a forward area with a ladder structure and uniform item locations.

Hall (1993) derived expressions for the expected travel time of a single trip, using the traversal strategy, which was initially studied by Kunder and Gudehus (1975). Assuming a simple ladder structure, Tang and Chew (1997) and

Chew and Tang (1999) studied the effects of different storage allocation strategies and batch sizes on the expected time to fill an order.

The shortest picking route for a single trip was examined for different types of forward areas. Ratliff and Rosenthal (1983) developed an efficient, DP-type procedure to determine the optimum picking route with a simple ladder structure. Algorithms to find the shortest picking route with multiple cross-aisles were developed by Roodbergen and De Koster (2001a, b) and Vaughan and Petersen (1999). The above papers also showed that adding cross-aisles would almost always reduce the total pick time. Daniels *et al.* (1998) considered the case where items are stored in multiple locations in the forward area.

A comparative study that considered many routing heuristics, and the interaction among the routing policies, shape of the forward area, and location of P/D points was performed by Petersen (1997), who also evaluated alternatives for making high-level picking policy decisions (Petersen 2000). De Koster and Van der Poort (1998) compared several heuristics against the optimal solution for routing pickers through a forward area. The effect of wide aisles on order picking routes was examined by Goetschalckx and Ratliff (1998). With wide aisles, the picker cannot access the racks on either side of an aisle without incurring additional travel.

Other research conducted on order batching has focused on automated warehouses or Automated Storage/Retrieval (AS/R) systems, which have a rectangular forward area and an S/R machine that follows the Chebyshev metric. Elsayed (1981) and Elsayed and Stern (1983) developed order batching algorithms to minimize the total distance travelled by the S/R machine. Elsayed and Unal (1989) also developed heuristics for the order-batching problem in an AS/R system as well as analytical models to estimate the S/R machine travel time. Won and Olafsson (2005) consider the joint order batching and order picking problems in an AS/R system in order to increase both warehouse efficiency and responsiveness. Additional studies to batch and schedule orders to minimize earliness and tardiness in an AS/R system were performed by Elsayed *et al.* (1993) and Elsayed and Lee (1996).

Pan and Liu (1995) present an overview and comparison of numerous batching policies for person-on-board AS/R systems, with the goal of minimizing total travel time. Malmberg and Al-Tassan (2000) present an analytical model to study the sensitivity of the performance of an AS/R system to system parameters and stock management policies.

Numerous authors studied the problem of locating and grouping items in a warehouse, often termed the stock location assignment problem. Liu (1999), Brynzer and Johansson (1996), and Rosenwein (1994) developed methods to group items together, which are often ordered together, in order to minimize total picker walking time. Assuming no demand dependency between the items, locating them based on the frequency of ordering was studied by Jarvis and McDowell (1991). Petersen (1999) and Petersen and Schmenner (1999) showed that for different routing policies, intelligent item location policies outperform random allocation; however, they did not account for congestion in the aisles. Caron *et al.* (1998) developed an efficient algorithm for assigning items to locations, and then developed analytical models for the expected travel distance required to pick the orders. The impact of item location in AS/R systems was examined by Malmberg and Deutsch (1988) for unit load systems, by Frazelle (1989) for miniload AS/R systems, and by Malmberg and Al-Tassan (1998) for less-than-unit-load systems. A number of

survey papers concerned with warehousing cite many of the above studies. The interested reader may refer to Cormier and Gunn (1992), Van den Berg (1999), and Rouwenhorts *et al.* (2000), among others.

Although research on the minisum order-batching problem has generated effective heuristics, the present paper examines special variants of the problem and it develops an MIP model. It also improves one of the heuristics, and using the MIP model, it presents a simple algorithm to determine a lower and an upper bound to the *optimal solution*, which has not been adequately addressed in the minisum order batching literature.

3. Order batching problem

This section discusses special cases of the problem to gain insight. It also introduces a Mixed-Integer Programming (MIP) model for the minisum order-batching problem. As shown below, the MIP model contains an approximation but is still useful. We first review our assumptions:

- When picking the orders on one trip, the picker follows the traversal strategy.
- The minimum-distance traversal route, which is straightforward to determine, is used when calculating the distance to pick each batch (or trip).
- The P/D point is located along the midpoint of the cross-aisle in the front. All trips start and end at the P/D point.
- The capacity of the pick cart/truck is expressed in items. No trip can exceed the capacity of the pick cart/truck.
- Each item consists of one piece. (It would be straightforward to relax this assumption).
- Order integrity is maintained. All items for a particular order must be picked on the same trip, i.e. an order cannot be split over two or more trips.
- Random storage is used. The items for an order are equally likely to be located anywhere in the forward area. Item location is unique; it is specified by the aisle number and position along the aisle.
- Travel is composed of in-the-aisle travel and cross-aisle travel. Additional vertical travel that may be incurred due to the height of the rack or height of an SKU is not considered. In addition, in-the-aisle travel is not affected by which side of the aisle an item is located.
- For simplicity, we assume one picker picks the entire wave. In reality, the resulting trips can be completed by multiple pickers. However, we do not address possible congestion in the aisles or interference among the pickers.
- Picker travel speed is constant throughout the forward area.
- The objective is to minimize the total distance required to pick all the orders in a wave (i.e. minisum).

We now address some of the special cases or variants of the problem.

3.1 Variants of the order-batching problem

The order-batching problem is a type of Clustered Traveling Salesman Problem (TSP) (Chisman 1975) or a Capacitated Vehicle Routing Problem (CVRP)

(Laporte 1992), with special properties due to the ladder structure. To begin the investigation, we first examine how problem difficulty is affected by the number of aisles, the order size (number of items), and variation in order size, which leads to the six variants of the problem shown in table 1. Note that if there is a single aisle (variants 1, 2 and 3), the picker does not traverse the full length of the aisle since he/she would simply return to the P/D point after picking the farthest item in the batch.

Variant 1: Optimal solution. For the first trip, the picker travels to the farthest order in the aisle, picks that order, and as he/she returns to the P/D point he/she continues to pick each order as they are encountered, until the cart capacity is reached. For the second trip, the picker travels to the farthest unpicked order and again picks each unpicked order encountered as he/she returns to the P/D point, and so on.

Proof: We focus on the number of trips and a lower bound (LB) on the length of each trip. Note that only the farthest order in each trip affects the total distance travelled for that trip, and each order has only one item. Therefore, ‘order’ and ‘item’ are interchangeable for variant 1. Consider first the number of trips needed. For a given wave of O orders, and a capacity of c items per trip, the minimum number of trips necessary, say, M is given by $\lceil O/c \rceil$. If a solution has m trips with $m > M$, we can take the trip with the shortest distance, and reassign its orders to other trips without changing the travel distance of the other trips. This would yield a solution with $m - 1$ trips, and a shorter overall travel distance, since we eliminated one trip and the distance of other trips remained the same. Hence, the optimal solution must have exactly M trips.

Consider next an LB on the travel distance for each of the M trips. Suppose the orders in the wave are numbered $\{1, 2, 3, \dots, O\}$ such that order 1 is the farthest order from the P/D point. That is, if $d(i)$ designates the distance travelled to order i , we number the orders such that $d(i) > d(i + 1)$ for all $i < O$. Without loss of generality, it is assumed that no two orders can have the same distance from the P/D point. Since all the orders must be picked, and since only the order with the farthest distance in each trip determines the trip length, some trip (which will be the longest trip) must contain order 1 and have a distance equal to $2d(1)$. Furthermore, given a capacity of c items, the second longest trip in any solution must have a distance greater than or equal to $2d(c + 1)$ since at most c items are picked on the longest trip. Hence, by induction, the minimum possible distance for the j th longest trip is equal to $2d((j - 1)*c + 1)$, regardless of how the other trips are constructed. Summing up the above minimum possible distance for each of the M trips yields an LB on the

Table 1. Order batching problem variants.

Variant	Number of aisles	Variation in order size	Order size
1	1	All equal	One item
2	1	All equal	Two or more items
3	1	Unequal	N/A
4	>1	All equal	One item
5	>1	All equal	Two or more items
6	>1	Unequal	N/A

distance required to pick all the orders. Since the algorithm we outlined results in a travel distance equal to the above LB, it yields the optimum solution. ■

Variant 2: Optimal solution. The same algorithm shown for variant 1 yields the optimum solution to variant 2. The only difference is that the ‘farthest order’ in the aisle is now defined as the order which contains the *farthest item* from the P/D point.

Proof: Since the items in an order must be picked on the same trip, all the items of order i can be ‘gathered’ at a point that overlaps with the *farthest item* of order i . This would not affect the total distance travelled to pick order i . Once the items for each order are gathered in the above fashion, the problem becomes equivalent to variant 1 simply by scaling the cart capacity. ■

Variant 3. Although variant 3 contains only one aisle, it is NP-hard.

Proof: Consider an instance of variant 3 where the farthest item on each order is located virtually at the same point in the aisle with negligible distance between them. For such an instance, the optimal solution is obtained by minimizing the total number of trips, which is the well-known bin-packing problem and known to be NP-complete (Garey and Johnson 1979). ■

Variant 4. The algorithm presented for variant 1 extends to variant 4 as follows. The picker travels to the farthest aisle that contains an unpicked order, picks the orders in that aisle, then travels towards the P/D point, entering aisles on a first-encountered-first-entered basis, picking orders in each aisle on a first-encountered-first-picked basis, without exceeding the cart capacity.

Unfortunately, the above algorithm does not necessarily yield the optimum solution for variant 4 as shown by Example 1 (table 2). The example has six aisles with the P/D point centred between aisles 3 and 4, and six orders, each with one item. Cart capacity is equal to three items. Assuming a cross-aisle distance of B units, and an aisle length of L units ($L > B$), the above algorithm, i.e. farthest aisle first, results in two trips and a total travel distance of $10B + 8L$ units. In contrast, the optimal solution has three trips (table 2) and a total travel distance of $12B + 6L$. Note that Example 1 also serves to show that the optimum bin packing solution, which is

Table 2. Example 1 for Variant 4.

Data			Solution		
Order number	Number of items	Aisle number with item	Technique	Travel distance	Trips
1	1	1	Farthest aisle first	$10B + 8L$	Trip 1: orders 1, 2, and 3
2	1	2			Trip 2: orders 4, 5, and 6
3	1	3			
4	1	4	Optimal	$12B + 6L$	Trip 1: orders 1 and 2
5	1	5			Trip 2: orders 3 and 4
6	1	6			Trip 3: orders 5 and 6

Table 3. Example 2 for Variant 5.

Data			Solution		
Order number	Number of items	Aisle number with item	Technique	Travel distance	Trips
1	2	1	Farthest aisle first	10B + 8L	Trip 1: orders 1, 2, and 3
2	2	2			Trip 2: orders 4, 5, and 6
3	2	3			
4	2	4	Optimal	12B + 6L	Trip 1: orders 1 and 2
5	2	5			Trip 2: orders 3 and 4
6	2	6			Trip 3: orders 5 and 6

clearly two trips, does not necessarily yield the optimum solution to the order batching problem, which has three trips in our example.

Variant 5. A similar example (see Example 2 in table 3) can be used to show that solving the bin-packing problem does not necessarily yield the optimum solution to variant 5. We simply add a second item (in the same aisle) to each order, and double the cart capacity to six items. As shown in table 3, assuming $B < L$, the solution with two trips has a total travel distance that is larger than the solution with three trips, which is the optimum solution for Example 2.

Variant 6. This variant represents the general case of the order-batching problem as we define it here, and it can be shown to be NP-hard. Consider an instance of variant 6, where all the items on all the orders are located at virtually the same point in the same aisle. For such an instance, the optimal solution is obtained by minimizing the total number of trips, which is the well-known bin-packing problem and known to be NP-complete (Garey and Johnson 1979).

It seems unlikely that a simple algorithm can be developed to find the optimal solution to variants 4 and 5; sophisticated algorithms or heuristics are likely to be needed to solve them. Variants 3 and 6, on the other hand, are NP-hard. Before we consider heuristic procedures, however, the next section discusses an MIP model for the order-batching problem.

3.2 MIP model for the order-batching problem

A conceptually straightforward way to model the order-batching problem is to model it as a set-partitioning problem (Balas and Padberg 1976). With such an approach, one would generate the ‘columns’, i.e. all possible trips that are feasible relative to cart capacity, and each ‘row’ would correspond to an order. Given the ‘cost’ of each column (i.e. travel distance), the objective would be to select a set of columns such that each order is covered exactly once, i.e. assigned to exactly one trip, while minimizing the total ‘cost’ or total distance travelled. Computing the ‘cost’ for each column (or trip) is straightforward. However, a shortcoming of the above approach is that for a problem of non-trivial size, the number of columns would be prohibitively large.

As discussed above, the order-batching problem is a type of Clustered TSP or capacitated VRP with special properties imposed by the ladder structure. In addition to the cart capacity, however, there is also order integrity, i.e. all the items of an order must be picked within the same trip. Although the second constraint is similar to the Clustered TSP, the simultaneous imposition of both constraints makes it impractical to solve the order-batching problem with traditional solutions developed for the TSP or VRP.

We therefore attempted to generate a compact, specialized model, but our efforts were generally not successful until we compromised and developed a model that solves the problem optimally only if the number of aisles visited on each trip in the final solution is an even number. If any trip contains an odd number of aisles, the in-the-aisle return distance for the *last aisle* visited on that trip is not captured in our model; however, all cross-aisle travel is captured regardless of the number of aisles visited on any trip. In other words, our model basically counts the number of aisles traversed on each trip, and it does not capture the ‘doubling-back’ that would occur in the last aisle of a trip if an odd number of aisles is visited. In reference to figure 1, note that if a trip visits an odd number of aisles, the picker must turn back in the last aisle in order to return to the P/D point. Of course, the picker can turn back after the last pick is performed (which would be the case in practice), but for analytical reasons it is assumed that all the aisles, *including the last one*, are traversed fully. In addition, it is assumed that the user specifies *a priori* the maximum number of trips that may be required to pick all the orders.

Due to the above limitations, we refer to the above model/problem as the ‘revised order batching model/problem’. However, despite its limitations, as shall be shown below, the solution to the revised order-batching problem provides very useful information for the original problem.

The *parameters* for the model are defined as follows:

- T maximum number of trips; specified by the user,
- A number of aisles,
- O number of orders in the wave,
- C capacity of the pick cart; expressed in the number of items,
- L length of an aisle; in distance units,
- s_i number of items in order i ($s_i \leq C$ for all i),
- r_i distance from the right-most item in order i to the P/D point. If the right-most item of order i is to the left of the P/D point, then we set $r_i = 0$,
- l_i distance from the left-most item in order i to the P/D point. If the left-most item of order i is to the right of the P/D point, then we set $l_i = 0$,
- $A_{ki} = 1$ if order i has a pick in aisle k ; 0 otherwise.

The *primary decision variable* for the model is Y_{ij} , which is equal to 1 if order i is assigned to trip j ; and 0 otherwise. The remaining decision variables depend on the value of Y_{ij} and are defined as follows:

- W_j length of trip j ,
- X_{kj} used to track if trip j visits aisle k ,
- d_j^r cross-aisle distance of trip j to the right of the P/D point,
- d_j^l cross-aisle distance of trip j to the left of the P/D point.

Given the above parameters and decision variables, the MIP model for the revised order batching problem is presented as follows:

$$\text{MIP : } \min \sum_{j=1}^T W_j \quad (1)$$

subject to:

$$\sum_{i=1}^O s_i Y_{ij} \leq C \quad \text{for all } j \quad (2)$$

$$\sum_{j=1}^T Y_{ij} = 1 \quad \text{for all } i \quad (3)$$

$$2(d_j^r + d_j^l) + L \sum_{k=1}^A X_{kj} = W_j \quad \text{for all } j \quad (4)$$

$$d_j^r \geq r_i Y_{ij} \quad \text{for all } i, j \quad (5)$$

$$d_j^l \geq l_i Y_{ij} \quad \text{for all } i, j \quad (6)$$

$$X_{kj} \geq A_{ki} Y_{ij} \quad \text{for all } i, j \text{ and } k, \text{ where } A_{ki} = 1 \quad (7)$$

$$W_j \geq W_{j+1} \quad \text{for } j = (1, 2, 3, \dots, T-1) \quad (8)$$

$$Y_{ij} \text{ binary } (0, 1) \quad (9)$$

Constraint (2) ensures that the cart capacity is not exceeded on any trip, while constraint (3) ensures that each order is assigned to exactly one trip. The total cross-aisle plus in-the-aisle distance travelled by trip j is calculated in constraint (4). Constraints (5) and (6), respectively, ensure that the appropriate right-most and left-most cross-aisle distance is computed for trip j . Constraint (7), which enters the model only when $A_{ki} = 1$, ensures that trip j visits aisle k if order i is assigned to trip j . Constraint (8), which at first may seem redundant, is a key constraint; it helps alleviate ‘symmetry’ when solving the MIP model. Note that, without constraint (8), the optimum set of orders to be picked in the first ‘batch’ can be assigned to any one of the T trips, the optimum set of orders in the second ‘batch’ can be assigned to any one of the $T-1$ trips, and so on, which would lead to many alternative optimum solutions, forcing any implicit search algorithm to discover all such solutions.

Of course, the size of any MIP model is of concern since the order-batching problem must be solved in a reasonable amount of time. In our model, only the decision variables designated by Y_{ij} need branching on; the other decision variables are all determined by the Y_{ij} values. Hence, the model size depends on the number of

Y_{ij} variables, which is determined by the total number of orders in the wave and the maximum number of trips allowed (T).

To use the model, the T value must be selected carefully; setting it ‘too high’ would unnecessarily increase the model size; setting it ‘too low’ could result in either a sub-optimal solution to the revised order batching problem or an infeasible solution space when the cart capacity constraint is considered. Suppose T^* is the number of trips in the optimum solution of the order batching problem, and T^{BP} is the number of trips obtained if the problem is solved (to optimality) strictly as a bin packing problem. Obviously, $T^{\text{BP}} \leq T^* \leq O$, which implies that we need to select T such that $T^{\text{BP}} \leq T \leq O$. We must also have $T \geq T^*$. That is, there must be a sufficient number of trips available to the MIP model; otherwise, the *best solution* found by the MIP model for T trips may actually be a sub-optimum solution to the revised problem. Of course, any trip that the model does not assign any orders to can be simply ignored.

If we set $T = \text{BP}^*$, it has two drawbacks: (1) one must first find the optimum bin packing solution, which itself is a difficult problem; and, more importantly (2) the resulting number of trips may be insufficient, that is, we may violate $T \geq T^*$. Therefore, a better strategy to pick a T value is first to solve the bin-packing problem heuristically, and then to add a few more trips. If the optimum solution to the MIP uses less than T trips, then we know that we have found the optimal solution to the revised order batching problem; however, if all T trips are used, we need to add another trip and rerun the MIP model.

3.3 Lower and upper bounds for the order-batching problem

As shown below, the optimum solution to the revised problem provides valuable information (lower and upper bounds) for the original order-batching problem.

Claim. The optimum solution to the revised order-batching problem is a lower bound for the original order-batching problem.

Proof: Let Z^* (Y_r^*) be the objective function value of the optimum solution to the original (revised) problem. Suppose we obtain Z_r^* from Z^* by subtracting L units from the total distance of each trip in Z^* that visits an *odd number* of aisles. Recall that L is the aisle length. Since $L > 0$, obviously $Z_r^* \leq Z^*$. (The inequality is not strict since it is possible that none of the trips in Z^* visit an odd number of aisles.) However, by construction, Z_r^* is an incumbent solution to the revised problem, which implies that $Y_r^* \leq Z_r^*$ since Y_r^* is the optimum solution. But $Z_r^* \leq Z^*$. Therefore, $Y_r^* \leq Z^*$ and the optimal solution to the revised problem is a lower bound for the original problem. ■

To determine the effectiveness of the above lower bound, one can compute the maximum difference between the optimal solution and the lower bound. The maximum difference would be $L(T_r^{0*})$, where (T_r^{0*}) is the number of trips (in the optimal solution to the revised problem) with an odd number of aisles. If all the trips in the optimum solution to the revised problem visit only an even number of aisles, and a sufficient number of trips have been provided, then the solution to the revised problem is also an optimum solution to the original order batching problem.

Given a solution to the revised problem, obtaining an *upper bound* for the original problem is straightforward. We simply add L units to the total distance of each trip in the revised solution that visits an odd number of aisles, which clearly leads to a feasible solution to the original problem and, therefore, represents an upper bound. Similar to the lower bound, to determine the effectiveness of the upper bound, one can compute the maximum difference between the optimal solution and the upper bound. The maximum difference would be $L(T_r^{o*})$, where (T_r^{o*}) is the number of trips (in the optimal solution to the revised problem) with an odd number of aisles. Since both the upper and/or lower bound could be equal to the optimal solution, the maximum difference between the bounds would also be $L(T_r^{o*})$. Note that the difference between the upper and lower bounds is maximized if all the trips in the optimal solution to the revised model visit an odd number of aisles.

The next section takes a closer look at recent heuristics developed for the original order-batching problem.

3.4 Heuristic approaches to the order-batching problem

Currently there are no mathematical programming-based models to solve optimally the minisum order-batching problem as defined in the present paper. However, several heuristics have been developed (e.g. Rosenwein 1996, Koster *et al.* 1999, Ruben and Jacobs 1999). Although we considered a number of heuristics for the present study, our numerical evaluations focus on only two of them since the results in the literature indicate that these two heuristics generally outperform their predecessors.

The first one is the ‘First Fit-Envelope Based Batching’ heuristic by Ruben and Jacobs (1999). In this heuristic, an ‘envelope index’ is calculated for each order based on the minimum and maximum aisle number visited by the order. Orders are then sorted by their envelope index, with ties broken in non-increasing order of the number of items in each order. With the orders sorted into a list, they are assigned to trips, by removing the orders, in sequence, from the above list and placing them in the lowest numbered batch with sufficient cart capacity to hold them. If no current trip has enough remaining capacity to hold the order, a new trip is started for that order. In the remainder of the paper, we refer to the above heuristic as the ‘R&J’ heuristic; the reader may refer to Ruben and Jacobs (1999) for details. The R&J heuristic is included in our study also since: it outperforms the strongest heuristic by Gibson and Sharp (1992); it was not tested against other heuristics evaluated in Koster *et al.* (1999); and it exploits the ladder structure of the forward area by considering the range, i.e. left-most and right-most aisle of each order, although this implies that it focuses primarily on minimizing cross-aisle travel while making no direct attempt to match orders that have common aisles *within* their ranges.

The second heuristic we focus on is a variant of the ‘C&W(ii)’ heuristic (Koster *et al.* 1999). The C&W(ii) heuristic is derived from Clarke and Wright’s work on routing vehicles from a central depot to a set of delivery points (Clarke and Wright 1964). The C&W(ii) heuristic, as presented by Koster *et al.* (1999), is a five-step procedure, where each order in the wave is considered a separate ‘cluster’. First, the savings (s_{ij}) for all possible pairs of clusters i and j (that do not exceed the cart capacity) are calculated as $s_{ij} = t_i + t_j - t_{ij}$, where t_i (t_j) and t_{ij} are the times to pick cluster i (j) alone and clusters i and j together, respectively. Second, the above pairs are sorted in a decreasing sequence of time saved. Third, the pair with the highest

savings is selected, with ties broken randomly. In the fourth step, the selected pair of clusters is combined to form a new cluster, if the cart capacity allows it. If not, the next pair on the list is selected and step four is repeated. The fifth step checks if any clusters can still be grouped together, given the cart capacity, and if so, it returns to step one to recalculate the time savings value for each pair of current clusters; otherwise, the heuristic terminates.

During the course of our research we observed that making an additional adjustment to the C&W(ii) heuristic, that is, using a normalized time savings value for each pair of orders, as opposed to the absolute time savings, can lead to improvements in the results. The only change needed to the above procedure is to calculate the savings for each pair as $s_{ij} = (t_i + t_j - t_{ij}) / (t_i + t_j)$. In the numerical results to follow, we use the C&W(ii) heuristic with the above adjustment based on normalized time savings. We refer to the adjusted C&W(ii) heuristic as ‘normalized time savings’ (NTS).

Before we present our numerical evaluations, we stress that a successful heuristic for the minisum order batching problem must take both cross-aisle and in-the-aisle travel into account. As we explained above, the R&J heuristic, for example, focuses on minimizing cross-aisle travel, which may perform well if there are many short aisles but is likely to weaken if there are fewer, longer aisles. One may define an alternative heuristic based on parts-clustering or part-family formation algorithms, where one seeks similarity between ‘parts’ (rows) based on the ‘processes’ (columns) they share. If the ‘rows’ and ‘columns’ represent the orders and the aisles, respectively, one can use a clustering algorithm to cluster the orders into trips based on the number of aisles they share. Such a heuristic should generally do well in minimizing in-the-aisle travel; however, it does not capture cross-aisle travel adequately.

Another point we demonstrated above is that, although the total distance travelled to pick all the orders in a wave depends in part on the number of trips performed, minimizing the number of trips is not sufficient to obtain a good or optimum solution. Hence, heuristics that are ‘bin packing oriented’ are not likely to perform consistently well either.

In contrast, the NTS heuristic attempts to capture both cross-aisle and in-the-aisle travel without limiting the number of trips performed. Therefore, one would expect it to be generally stronger than other heuristics and less sensitive to the configuration/shape of the forward area. However, since previous studies did not obtain an optimum solution or a lower bound to the order batching problem, the performance of even the strongest heuristic relative to the optimum solution is generally not known. In the next section, we numerically compare the performances of the R&J and the NTS heuristics. We also attempt to identify any ‘performance gaps’ or ‘opportunities’ relative to the optimum solution.

4. Numerical evaluation of the heuristics

Most RDCs face variable order sizes (expressed in number of items/order) as discussed by Ruben and Jacobs (1999). Therefore, for the numerical evaluation, it is assumed that orders come in various sizes, which will be referred to as the ‘order type’. To generate the orders, we first determine the total number of orders in the wave (O), the number of order types (K), the probability of finding order type K in

the wave (p_K), and the minimum (m_K^-) and maximum (m_K^+) number of items in order type K . Given the above values, each order is generated independently by first randomly selecting the order type according to the specified probabilities. Once the order type is known, the number of items is determined by randomly and uniformly selecting an integer in the interval $[m_K^-, m_K^+]$. Given the number of items for the order, the location of each item is determined by the aisle number and the distance from the front of the aisle. Since a uniform distribution is used, each item is equally likely to be in any aisle, and the distance from the front of the aisle is a uniform random variable over the length of the aisle (L). The capacity of the pick cart (C) is expressed in number of items.

For practical reasons, all the aisles are assumed to have the same length, and the distance between any pair of adjacent aisles, D_b , is assumed to be equal. To allow for turning clearance when entering and exiting an aisle, an in/out distance, $D_{i/o}$, is added for each aisle visited. Since $D_{i/o}$ accounts for both the in and out distance, it is added only once for each aisle entered by the picker. The additional time required to pick an item from the rack is not included in the results since it does not depend on the picking route.

4.1 Parameter values

We use the following parameter values in our numerical evaluations:

- Number of orders per wave: $O = (20, 25)$,
- Number of order types: $K = 4$,
- Probability of each order type: $p_K = (0.25, 0.35, 0.25, 0.15)$,
- Minimum value for each order type: $m_K^- = (1, 3, 6, 9)$,
- Maximum value for each order type: $m_K^+ = (2, 5, 8, 10)$,
- Pick cart capacity: $C = 25$ items,
- Distance between aisles: $D_b = 10$ feet; in/out distance: $D_{i/o} = 5$ feet,
- Number of aisles: $A = 11, 16, 22$ aisles for 20 orders, and $A = 16$ aisles for 25 orders,
- Aisle length (L) for each A value above: $L = 227.27, 156.25, 113.65$ feet,
- Maximum number of trips: $T = (6, 8)$.

The above parameters yield an average order size of 4.95 items/order or approximately 5 items/order. Three different shapes or configurations for the forward area are used; in each configuration there are 5000 linear feet of aisle space. The range of data and parameter values we consider for the study do not reflect a single RDC. Rather, they are based on our observations of multiple RDCs. While we tried to pick parameter values that are typical or at least would cover a range of applications, specific parameter values can still vary depending on the specific application such as distributing music CDs/DVDs versus groceries.

4.2 Numerical results

The results presented in tables 4–7, where ‘Dist.’ under ‘MIP’ represents the lower bound (LB), and ‘Dist.’ under ‘MIP + odd if needed’ represents the upper bound (UB), indicate that the NTS heuristic outperforms the R&J heuristic by 12% in overall pick time. However, on average, NTS is about 11% above the LB,

and 6.0–7.5% above the UB. Recall that the two bounds are obtained through the MIP model. Furthermore, although NTS outperforms R&J in overall pick time, it runs substantially slower. In fact, on a ‘typical’ desktop computer, R&J runs almost instantaneously, while NTS can take several minutes to solve larger problems.

Of course, when $LB = UB$, the solution to the revised model is an optimal solution to the order batching problem. This occurred in problem 17 in table 4, and in problems 4 and 13 in table 6. One drawback of the MIP model is that if the solution contains as many trips as the maximum number of trips, then the solution to the revised model does not constitute an LB because under some conditions, as illustrated above in Example 1 (table 2), the optimal solution contains more than the minimum number of trips needed to pick all the orders. Therefore, the LBs for problem 18 in table 4, problem 12 in table 5, problems 5 and 18 in table 6, and problem 6 in table 7 are not necessarily valid LBs for the original order batching problem. However, they are valid LBs if one limits the maximum number of trips allowed.

The results in tables 4–7 suggest that there is room for improvement in current heuristics developed for order batching. To explore this opportunity, we changed the tie-breaking strategy in the NTS heuristic by selecting the largest combined order size as in Rosenwein (1996). Another possible change is to select the pair of orders with the largest combined order size in the top 10% of the total time savings list. Neither change, however, led to improvements that are worth reporting here.

The size of the order batching problem, i.e. the number of orders/wave or ‘wave size’, is the main difficulty. Although the MIP model developed provides lower and upper bounds, it would require substantial computational time for large waves. Therefore, for large, real-world problems, we still need heuristic procedures to batch the orders; however, the choice is not simple. For 200 orders, the NTS heuristic can take almost 7 h to run on a desktop computer. While R&J runs under 1 min, NTS yields travel time savings of about 18% over R&J even for waves with 25 or less orders (tables 4–7). A compromise may be found in the C&W(i) heuristic (Koster *et al.* 1999), which is identical to C&W(ii) except that the time savings list is not recalculated after each pair of orders is grouped. This reduces the run time substantially but at the expense of overall performance. As before, we used normalized time savings when implementing C&W(i). For 200 orders, for example, the run time of C&W(i) is approximately one-tenth of that of NTS. However, C&W(i) outperforms R&J only by 5%.

Hence, from a research standpoint, the results imply that there is an opportunity to develop a better order batching heuristic but the runtime is going to be critical. From a practical or managerial perspective, for waves with 25 or so orders, the NTS heuristic runs reasonably fast and the solutions are reasonably good, although they are inconsistent in quality as indicated by the minimum and maximum ‘percent above UB’ values shown in tables 4–7. For larger waves, NTS will require too much time and, therefore, the user must consider less effective but faster heuristics such as R&J or C&W(i). Of course, with desktop computers gaining more speed over time, NTS may become viable for many applications in the near future.

The wave size is another key decision from an operational and managerial perspective. Not considering its impact on subsequent operations such as packing or shipping, a larger wave would generally yield shorter expected distances travelled per order picked, provided that good batches (trips) can be constructed regardless of

Table 4. Twenty orders, 11 Aisles.

MIP				MIP + odd if needed				NTS (Relative Values)				R&J		R&J vs NTS		
Prob. #	Dist.	Tours	Aisles	added	Dist.	Tours	% Above LB	Dist.	Tours	% Above LB	% Above UB	Dist.	Tours	% Difference		
1	7395.83	4		1	7628.10	4	3.14%	8737.18	5	18.14%	14.54%	8617.18	4	-1.37%		
2	7920.37	4		3	8617.18	4	8.80%	8617.18	4	8.80%	0.00%	9546.26	4	10.78%		
3	9938.53	5		1	10170.80	5	2.34%	10170.80	5	2.34%	0.00%	11584.42	5	13.90%		
4	7920.37	4		3	8617.18	4	8.80%	8637.18	5	9.05%	0.23%	10070.80	4	16.60%		
5	7940.37	4		1	8172.64	4	2.93%	9101.72	4	14.63%	11.37%	10495.34	4	15.31%		
6	9938.53	5		1	10170.80	5	2.34%	10695.34	5	7.61%	5.16%	12048.96	5	12.66%		
7	5922.21	4		3	6619.02	4	11.77%	7203.56	4	21.64%	8.83%	8112.64	4	12.62%		
8	7668.10	4		2	8132.64	4	6.06%	8737.18	5	13.94%	7.43%	11604.42	5	32.82%		
9	9686.26	5		4	10615.34	5	9.59%	10230.80	5	5.62%	-3.62%	12088.96	5	18.16%		
10	10190.80	5		2	10655.34	5	4.56%	10675.34	5	4.75%	0.19%	12068.96	5	13.05%		
11	10230.80	5		2	10695.34	5	4.54%	12228.96	6	19.53%	14.34%	11624.42	5	-4.94%		
12	9121.72	5		2	9586.26	5	5.09%	10190.80	5	11.72%	6.31%	11544.42	5	13.28%		
13	8717.18	5		2	9181.72	5	5.33%	9706.26	5	11.35%	5.71%	11584.42	5	19.35%		
14	9081.72	4		2	9546.26	4	5.12%	10150.80	5	11.77%	6.33%	10070.80	4	-0.79%		
15	6174.48	4		2	6639.02	4	7.52%	7203.56	4	16.67%	8.50%	8597.18	4	19.35%		
16	10250.80	5		2	10715.34	5	4.53%	11299.88	6	10.23%	5.46%	13562.58	6	20.02%		
17	7728.10	4		0	7728.10	4	0.00%	9301.72	5	20.36%	20.36%	10030.80	4	7.84%		
18	11067.61	6		3	11764.42	6	6.30%	12228.96	6	10.49%	3.95%	14571.66	6	19.16%		
19	9221.72	5		2	9686.26	5	5.04%	10655.34	5	15.55%	10.00%	12088.96	5	13.45%		
20	9141.72	5		2	9606.26	5	5.08%	10655.34	6	16.56%	10.92%	12513.50	5	17.44%		
Average																
Minimum															13.43%	
Maximum															-4.94%	
															32.82%	
Number of orders = 20															Number of Aisles = 11	
Capacity = 25															Aisle length = 227.27	
Max tours = 6															In out distance = 5	
Number of Binary Variables = 120															Between Aisles = 10	

wave size. In fact, comparing the UBs in tables 5 and 7 with 16 aisles, increasing the wave size from 20 to 25 orders decreases the average distance/order from 429.21 to 397.47 feet/order or by about 7.5%. However, large waves are more difficult to optimize, and they may also cause delays since incoming orders would have to wait longer before they can be picked. Clearly, this is a trade-off that needs further investigation.

5. Conclusions

A Mixed-Integer Programming (MIP) model was developed to determine lower and upper bounds to the minimum order batching problem in a ladder structure walk-and-pick forward area, where each aisle entered by the picker is traversed fully. Using the bounds, we have shown that current heuristic methods are reasonably good, although there is still room for improvement. Unfortunately, the best batching heuristics that yield solutions with the least walk times require substantial computational time for large waves.

Possible focus areas for further research in order batching include: reducing the time required to solve the MIP; reducing the computational time of the best heuristics or improving the solutions obtained from the fast heuristics; and determining ideal wave sizes based on total throughput time through the system.

References

- Balas, E. and Padberg, M.W., Set partitioning: a survey. *SIAM Rev.*, 1976, **18**(4), 710–760.
- Brynzner, H. and Johansson, M.I., Storage location assignment: using the product structure to reduce order picking times. *Int. J. Prod. Econ.*, 1996, **46**, 595–603.
- Caron, F., Marchet, G. and Perego, A., Routing policies and COI-based storage policies in picker-to-part systems. *Int. J. Prod. Res.*, 1998, **36**(3), 713–732.
- Chew, E.P. and Tang, L.C., Travel time analysis for general item location assignment in a rectangular warehouse. *Eur. J. Oper. Res.*, 1999, **112**(3), 582–597.
- Chisman, J.A., The Clustered Traveling Salesman Problem. *Comput. Oper. Res.*, 1975, **2**, 115–119.
- Clarke, G. and Wright, J.W., Scheduling of vehicles from a central depot to a number of delivery points. *Oper. Res.*, 1964, **12**(4), 568–581.
- Cormier, G. and Gunn, E.A., A review of warehouse models. *Eur. J. Oper. Res.*, 1992, **58**(1), 3–13.
- Daniels, R.L., Rummel, J.L. and Schantz, R., A model for warehouse order picking. *Eur. J. Oper. Res.*, 1998, **105**(1), 1–17.
- De Koster, R. and Van der Poort, E., Routing orderpickers in a warehouse: a comparison between optimal and heuristic solutions. *IIE Trans.*, 1998, **30**(5), 469–480.
- Elsayed, E.A., Algorithms for optimal material handling in automatic warehousing systems. *Int. J. Prod. Res.*, 1981, **19**(5), 525–535.
- Elsayed, E.A. and Lee, M.K., Order processing in automated storage/retrieval systems with due dates. *IIE Trans.*, 1996, **28**(7), 567–577.
- Elsayed, E.A., Lee, M.K., Kim, S. and Scherer, E., Sequencing and batching procedures for minimizing earliness and tardiness penalty of order retrievals. *Int. J. Prod. Res.*, 1993, **31**(3), 727–738.
- Elsayed, E.A. and Stern, R.G., Computerized algorithms for order processing in automated warehousing systems. *Int. J. Prod. Res.*, 1983, **21**(4), 579–586.

- Elsayed, E.A. and Unal, O.I., Order batching algorithms and travel-time estimation for automated storage/retrieval systems. *Int. J. Prod. Res.*, 1989, **27**(7), 1097–1114.
- Frazelle, E.H., Stock location assignment and order picking productivity. Doctoral dissertation, Georgia Institute of Technology, Atlanta, GA, 1989.
- Gademann, A.J.R.M., Van den Berg, J.P. and Van der Hoff, H.H., An order batching algorithm for wave picking in a parallel-aisle warehouse. *IIE Trans.*, 2001, **33**(5), 385–398.
- Garey, M.R. and Johnson, D.S., Computers and Intractability: A Guide to the Theory of NP-Completeness, 1979 (W.H. Freeman and Company: New York).
- Gibson, D.R. and Sharp, G.P., Order batching procedures. *Eur. J. Oper. Res.*, 1992, **58**(1), 57–67.
- Goetschalckx, M. and Ratliff, H.D., Order picking in an aisle. *IIE Trans.*, 1998, **20**(1), 53–62.
- Hall, R.W., Distance approximations for routing manual pickers in a warehouse. *IIE Trans.*, 1993, **25**(4), 76–87.
- Jarvis, J.M. and McDowell, E.D., Optimal product layout in an order picking warehouse. *IIE Trans.*, 1991, **23**(1), 93–102.
- Koster, M.B.M., Van der Poort, E.S. and Wolter, M., Efficient orderbatching methods in warehouses. *Int. J. Prod. Res.*, 1999, **37**(7), 1479–1504.
- Kunder, R. and Gudehus, T., Mitlere Wegzeiten beim eindimensionalen Kommissioniere. *Zeitschrift fur Oper. Res.*, 1975, **19**, B53–B72.
- Laporte, G., The vehicle routing problem: an overview of exact and approximate algorithms. *Eur. J. Oper. Res.*, 1992, **59**, 345–358.
- Liu, C.M., Clustering techniques for stock location and order-picking in a distribution center. *Comput. Oper. Res.*, 1999, **26**(10–11), 989–1002.
- Malmberg, C.J. and Al-Tassan, K., An integrated performance model for orderpicking systems with randomized storage. *Appl. Math. Model.*, 2000, **24**(2), 95–111.
- Malmberg, C.J. and Al-Tassan, K.M., Analysis of storage assignment policies in less than unit load warehousing systems. *Int. J. Prod. Res.*, 1998, **36**(12), 3459–3475.
- Malmberg, C.J. and Deutsch, S.J., A stock location model for dual address order picking systems. *IIE Trans.*, 1988, **20**(1), 44–52.
- Pan, C.H. and Liu, S.Y., A comparative study of order batching algorithms. *Omega Int. J. Manag. Sci.*, 1995, **23**(6), 691–700.
- Petersen II, C.G. II, An evaluation of order picking routeing policies. *Int. J. Oper. Prod. Manag.*, 1997, **17**(11), 1098–1111.
- Petersen II, C.G. II, The impact of routing and storage policies on warehouse efficiency. *Int. J. Oper. Prod. Manag.*, 1999, **19**(10), 1053–1064.
- Petersen II, C.G. II, An evaluation of order picking policies for mail order companies. *Prod. Oper. Manag.*, 2000, **9**(4), 319–335.
- Petersen II, C.G. II and Schmenner, R.W., An evaluation of routing and volume-based storage policies in an order picking operation. *Dec. Sci.*, 1999, **30**(2), 481–501.
- Ratliff, H.D. and Rosenthal, A.S., Order-picking in a rectangular warehouse: a solvable case of the Traveling Salesman Problem. *Oper. Res.*, 1983, **31**(3), 507–521.
- Roodbergen, K.J. and de Koster, R., Routing methods for warehouses with multiple cross aisles. *Int. J. Prod. Res.*, 2001b, **39**(9), 1865–1883.
- Roodbergen, K.J. and de Koster, R., Routing order pickers in a warehouse with a middle aisle. *Eur. J. Oper. Res.*, 2001a, **133**(1), 32–43.
- Rosenwein, M.B., A comparison of heuristics for the problem of batching orders for warehouse selection. *Int. J. Prod. Res.*, 1996, **34**(3), 657–664.
- Rosenwein, M.B., An application of cluster analysis to the problem of locating items within a warehouse. *IIE Trans.*, 1994, **26**(1), 101–103.
- Rouwenhorts, B., Reuter, B., Stockrahm, V., Van Houtum, G.J., Mantel, R.J. and Zijm, W.H.M., Warehouse design and control: framework and literature review. *Eur. J. Oper. Res.*, 2000, **122**(3), 515–533.
- Ruben, R.A. and Jacobs, F.R., Batch construction and storage assignment. *Manag. Sci.*, 1999, **45**(4), 575–596.
- Tang, L.C. and Chew, E.P., Order picking systems: batching and storage assignment strategies. *Comput. Ind. Eng.*, 1997, **33**(3–4), 817–820.

- Tompkins, J.A., White, J.A., Bozer, Y.A. and Tanchoco, J.M.A., *Facilities Planning*, 3rd ed., 2003 (Wiley: New Jersey).
- Van den Berg, J.P., A literature survey on planning and control of warehousing systems. *IIE Trans.*, 1999, **31**(8), 751–762.
- Vaughan, T.S. and Petersen, C.G., The effect of warehouse cross aisles on order picking efficiency. *Int. J. Prod. Res.*, 1999, **37**(4), 881–897.
- Won, J. and Olafsson, S., Joint order batching and order picking in warehouse operations. *Int. J. Prod. Res.*, 2005, **43**(7), 1427–1442.